MACHINE WHITE PAPERS

Samsung SL-45A/2200 — Hammer of Forge

Heavy CNC lathe purpose-built for forged blanks, rough shafts, drive mandrels, and structural stock. 14" OD × 96" L max turning.



Hammer of Forge · heavy-duty CNC lathe for rough stock

Hammer of Forge is the machine that takes ugly starting material and makes it round, straight, and to size. Forged blanks with scale and skin, rough saw-cut billets, oversized bar stock — this machine has the rigidity and horsepower to get through the roughing pass without chatter and without walking the part. Most of what our other lathes finish, this one starts.

What Hammer of Forge is for

Hammer of Forge is the machine that takes ugly starting material and makes it round, straight, and to size. Forged blanks still carrying scale and skin, rough saw-cut billets, oversized bar — 14" OD × 96" L, a 6.5" bar capacity, and a 3-speed gearbox that puts real torque at the bottom end. Most of what the other lathes in this shop finish, this one starts. Roughing forged stock is a rigidity and torque problem, and it is genuinely hard on a machine that wasn't built for it.

Pick this machine when

- The starting material is a forging or casting with scale, skin, or an interrupted first cut.

- You need heavy stock removal before a finish operation, and you'd rather not spend a precision machine's life doing it.
- Long rough shafts, drive mandrels, and structural stock up to 96".
- Bar work up to 6.5" through the spindle.

> **When it's the wrong machine**
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> Finish-critical work as a one-hit. This machine's job is to get material off predictably, and the honest sequence on a tight-tolerance part is rough here, finish elsewhere. Small precision parts are a poor fit for the same reason. If the part starts as clean bar and finishes to ±0.0005" in one operation, a Hi-Eco cell is the better machine.

From the floor

Scale is abrasive and the first cut through it is an interrupted cut into an unknown surface — that combination eats inserts and, worse, eats them unpredictably. The first pass gets taken deep enough to get under the scale in one go rather than skimming it, because a tool tip dragging along the scale layer wears far faster than one cutting clean material beneath it.

Pick your industry

Samsung SL-45A for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

Samsung SL-45A for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

Samsung SL-45A for Military & Defense

Full spec sheet, typical defense parts, alloys, tolerance expectations, documentation approach.

Samsung SL-45A for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys, tolerance expectations, documentation approach.

Frequently asked

What does B&R Productions use the Samsung SL-45A/2200 for?

When is the Hammer of Forge the right machine for my part?

When is the Hammer of Forge the wrong machine?

What industries does B&R machine on the Samsung SL-45A?

How do I get a quote for work on the Samsung SL-45A?

Have a project?